



Polynomials and Arithmetic

Polynomials can be added, subtracted, multiplied, and divided just like real numbers can. When you add, subtract, or multiply polynomials, you always end up with a polynomial result, which is not always true when dividing polynomials.

Closure

Definition

A set is **closed under an operation** if the result of the operation on any two elements of the set is also an element of the set.

For instance, the set of even numbers is closed under addition. If you add any two even numbers, the sum is also an even number.

EXAMPLE 1

Determine whether the set is closed under subtraction. If so, explain why; if it is not closed, give a counterexample.

A the set of positive integers

SOLUTION

The set of positive integers is not closed under subtraction. For instance, $6 - 9 = -3$ and -3 is not a positive integer.

B the set of even integers

SOLUTION

The set of even integers is closed under subtraction. An even integer can be written as $2x$, where x is an integer. Subtract two even integers $2p$ and $2q$: $2p - 2q = 2(p - q)$. The value $(p - q)$ is an integer, so $2 \cdot (p - q)$ is an integer as well. The set of even integers is closed under subtraction. ■

► **Think About It** A **counterexample** is an example that shows that a statement is false. You need only a single counterexample to show that a statement is false.

Operations Under Which Polynomials Are Closed

The set of polynomials is closed under addition, subtraction, and multiplication.

► **Remember** A polynomial is a monomial or the sum of monomials. The variables in polynomials must have whole-number exponents. Also, a constant without a variable, such as -4 , is considered a polynomial because it is a monomial.

EXAMPLE 2

Explain why the set of polynomials is closed under addition.

SOLUTION

Start with a simple case—the addition of two binomials of degree one.

$(ax + b) + (cx + d)$ where a , b , c , and d are all integers

$$\begin{aligned}(ax + b) + (cx + d) &= (ax + cx) + (b + d) \\ &= (a + c)x + (b + d)\end{aligned}$$

The values $(a + c)$ and $(b + d)$ are integers, so the result of adding two binomials, $(ax + b) + (cx + d)$, is also a polynomial.

Expanding this argument, you can use like terms to simplify the sum of any polynomials. Consider the sum of the polynomials.

$$(ax^3 + bx^2 + cx + d) + (px^3 + qx^2 + rx + s)$$

The result of this sum is a polynomial.

$$(a + p)x^3 + (b + q)x^2 + (c + r)x + (d + s) \blacksquare$$

► **Remember** The degree of a monomial is the sum of the exponents of its variable factors. The degree of a polynomial is the degree of the monomial with the greatest degree.

Operations Under Which Polynomials Are Not Closed

EXAMPLE 3

Give a counterexample to show that polynomials are not closed under division.

SOLUTION

There are many instances in which polynomials are not closed under division. For instance, consider $(x - 4) \div (x^2 - 4x)$.

$$\frac{x - 4}{x^2 - 4x} = \frac{x - 4}{x(x - 4)} = \frac{1}{x}$$

A monomial cannot have a variable in a denominator, so $\frac{1}{x}$ is not a polynomial. $\blacksquare$